

| Tool Version : Vivado v.2025.1 (win64) Build
6140274 Thu May 22 00:12:29 MDT 2025
| Date : Mon Apr 13 17:37:14 2026
| Host : DESKTOP-STEMT34 running 64-bit
major release (build 9200)
| Command : report_power -file
top_power_routed.rpt -pb top_power_summary_routed.pb
-rpx top_power_routed.rpx
| Design : top
| Device : xc7a100tcsg324-1
| Design State : routed
| Grade : commercial
| Process : typical
Characterization : Production

Power Report

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1. Summary

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Total On-Chip Power (W)	0.145	
Design Power Budget (W)	Unspecified*	
Power Budget Margin (W)	NA	
Dynamic (W)	0.048	
Device Static (W)	0.097	
Effective TJA (C/W)	4.6	
Max Ambient (C)	84.3	
Junction Temperature (C)	25.7	
Confidence Level	Low	
Setting File	---	
Simulation Activity File	---	
Design Nets Matched	NA	
+-----+-----+		

* Specify Design Power Budget using,
set_operating_conditions -design_power_budget <value in
Watts>

1.1 On-Chip Components

On-Chip		Power (W)	Used
Available	Utilization (%)		
Clocks		<0.001	3
---	---		
Slice Logic		0.010	1347
---	---		
LUT as Logic		0.009	590
63400	0.93		
Register		<0.001	539
126800	0.43		
LUT as Distributed RAM		<0.001	32
19000	0.17		
F7/F8 Muxes		<0.001	97
63400	0.15		
CARRY4		<0.001	39
15850	0.25		
Others		0.000	10
---	---		
BUFG		0.000	1
32	3.13		
Signals		0.015	1188
---	---		
I/O		0.022	15
210	7.14		
Static Power		0.097	

Total	0.145

1.2 Power Supply Summary

Source	Voltage (V)	Total (A)	Dynamic (A)
Static (A)	Powerup (A)	Budget (A)	Margin (A)
Vccint	1.000	0.041	0.026
0.015	NA	Unspecified	NA
Vccaux	1.800	0.019	0.001
0.018	NA	Unspecified	NA
Vcco33	3.300	0.010	0.006
0.004	NA	Unspecified	NA
Vcco25	2.500	0.000	0.000
0.000	NA	Unspecified	NA
Vcco18	1.800	0.000	0.000
0.000	NA	Unspecified	NA
Vcco15	1.500	0.000	0.000
0.000	NA	Unspecified	NA
Vcco135	1.350	0.000	0.000
0.000	NA	Unspecified	NA
Vcco12	1.200	0.000	0.000
0.000	NA	Unspecified	NA

Vccaux_io	1.800	0.000	0.000
0.000	NA	Unspecified	NA
Vccbram	1.000	0.000	0.000
0.000	NA	Unspecified	NA
MGTAVcc	1.000	0.000	0.000
0.000	NA	Unspecified	NA
MGTAVtt	1.200	0.000	0.000
0.000	NA	Unspecified	NA
Vccadc	1.800	0.020	0.000
0.020	NA	Unspecified	NA

1.3 Confidence Level

User Input Data	Confidence	Details
		Action
Design implementation state	High	Design is
routed		

Clock nodes activity	Low	User	
specified less than 75% of clocks			
Provide missing clock activity with a constraint file,			
simulation results or by editing the "By Clock Domain"			
view			
I/O nodes activity	Low	More than	
75% of inputs are missing user specification		Provide	
missing input activity with simulation results or by			
editing the "By Resource Type -> I/Os" view			
Internal nodes activity	Medium	User	
specified less than 25% of internal nodes			
Provide missing internal nodes activity with simulation			
results or by editing the "By Resource Type" views			
Device models	High	Device	
models are Production			
Overall confidence level	Low		
+-----+-----+-----+			
-----+-----			

2. Settings

2.1 Environment

Ambient Temp (C)	25.0
ThetaJA (C/W)	4.6
Airflow (LFM)	250
Heat Sink	medium (Medium Profile)
ThetaSA (C/W)	4.6
Board Selection	medium (10"x10")
# of Board Layers	12to15 (12 to 15 Layers)
Board Temperature (C)	25.0

2.2 Clock Constraints

Clock	Domain	Constraint (ns)
sys_clk_pin	CLK100MHZ	10.0

3. Detailed Reports

3.1 By Hierarchy

+-----+-----+	
Name	Power (W)
+-----+-----+	
top	0.048
cpu	0.025
dp	0.025
pcreg	0.007
rf	0.017
+-----+-----+	